Math 205A - Fall 2014
Homework # 1
Solutions

Problem 1: Show that f is well-defined, monotone, continuous and that it is constant on every interval contained in the complement of the Cantor set.

If x has two ternary expansions $0.a_1a_2\ldots$ and $0, a_1'a_2'\ldots$, then one of them has a tail of $2s$, say $0.a_1a_2\ldots$. In this case, there $\exists m$ such that $a_n = a_n', \forall n \leq m$, a_{m+1} is 0 or 1, and $a_{m+2} = a_{m+3} = \ldots = 2$, whereas $a'_{m+1} = a_{m+1} + 1$, and $a'_{m+2} = a'_{m+3} = \ldots = 0$. Denote by N, N' and b_n, b'_n respectively the quantities of the problem. Note first that $b_n = b'_n$ for $n \leq m$. So if $N \leq m$: $b_n = b'_n$ for all n. Otherwise:
If $a_{m+1} = 0$: $a'_{m+1} = 1$, $N = \infty$ and $N' = m + 1$, then

$$\sum_1^N \frac{b_n}{2^n} = \sum_1^m \frac{b_n}{2^n} + \frac{0}{2^{m+1}} + \sum_{m+2}^{\infty} \frac{1}{2^n} = \sum_1^m \frac{b_n}{2^n} + \frac{1}{2^{m+1}}, \qquad \sum_{n=1}^{N'} \frac{b'_n}{2^n} = \sum_{n=1}^m \frac{b_n}{2^n} + \frac{1}{2^{m+1}}.$$

If $a_{m+1} = 1$: $a'_{m+1} = 2$, $N = m + 1$ and $N' = \infty$, then

$$\sum_1^N \frac{b_n}{2^n} = \sum_1^m \frac{b_n}{2^n} + \frac{1}{2^{m+1}}, \qquad \sum_1^{N'} \frac{b'_n}{2^n} = \sum_1^m \frac{b_n}{2^n} + \frac{1}{2^{m+1}} + \sum_{m+2}^{\infty} \frac{0}{2^n} = \sum_1^m \frac{b_n}{2^n} + \frac{1}{2^{m+1}}.$$

In all cases the results coincide. Hence f is well-defined.

Now let $R_n(x)$ be 1 if the interval removed during the nth step of the construction of the Cantor set lies before x, and 1 otherwise. Then $f(x) = \sum R_n(x)2^{-n}$. Since each $R_n(x)$ is monotone increasing in x, $f(x)$ is monotone too.

Given an interval I in C^c, each removed interval other then I lies either before, or after I, so all the corresponding $R_n(x)$'s coincide in I. When I is removed, that R_n is 1 for all $x \in I$. Hence, $f(x)$ is constant on I.

f is onto, because given $\sum b_n 2^{-n}$ in $[0, 1]$, it's preimage contains at least $\sum(2 \cdot b_n)3^{-n}$. So $f : [0, 1] \longrightarrow [0, 1]$ is monotone and onto, hence continuous. □

Problem 2: Show that g is a homeomorphism of $[0, 1]$ onto $[0, 2]$ and that $m[g(C)] = 1$. Show that there exists a measurable set A so that $p^{-1}(A)$ is not measurable. Show that there is a measurable set that is not a Borel set.

g is continuous, strictly increasing (hence 1-1) and $g(0) = 0$, $g(1) = 2$, so onto. Thus g^{-1} exists. g^{-1} is also strictly increasing and onto, hence continuous.

C^c is a union of disjoint intervals I_j with $\sum |I_j| = m(C^c) = 1$, and f is constant on each of them. On an interval I_j, $g(x) = f(x_j) + x$, so $m(g(I)) = m(I_j + f(x_j)) = |I_j|$. g being a homeomorphism implies $m(g(C^c)) = \sum m(g(I_j)) = \sum |I_j| = m(C^c) = 1$, hence $m(g(C)) = 1$.

Next we prove that every measurable subset $E \subseteq [0, 2]$ with $m(E) > 0$ has a non-measurable subset: Let $P \in [0, 1)$ be the non-measurable set constructed by picking exactly one element of each of the equivalence classes of the relation $x \sim y \iff x - y \in \mathbb{Q}$. Let also $P_q = P + q$ with addition mod 2, and $E_q = E \cap P_q$. Then $[0, 2] = \cup_{q\in\mathbb{Q}\cap[0,2]} P_q$, hence $E = \cup_{q\in\mathbb{Q}\cap[0,2]} E_q$, with a disjoint union on the right (same proof as in class). We claim that E_q is non-measurable for some q: in deed if not, $E_q + r$'s are disjoint for $r \in \mathbb{Q}\cap[0, 2]$, so $\sum_{r\in\mathbb{Q}\cap[0,2]} m(E_q + r) = m(\cup_{r\in\mathbb{Q}\cap[0,2]} E_q + r) \leq m([0, 2]) = 2$, so $m(E_q) = 0$ for all q. But then $m(E) = m(\sum_{q\in\mathbb{Q}\cap[0,2]} E_q) = 0$, a contradicting the positivity of $m(E)$. Hence, some $E_q \subset E$ is non-measurable.

Now let $E \subset g(C)$ be non-measurable, and let $A = p(E)$. A is measurable since $A \subset C$ and $m(C) = 0$, but $p^{-1}(A) = E$ is not. This A is also an example of a measurable but not Borel set, because if it was Borel, then since p is continuous, so would be $p_{-1}(A) = E$.

□

Problem 3:
(i) Lower semicontinuous if and only if $\{x : f(x) > \lambda\}$ open for all λ.
(ii) If f, g lower semicontinuous at y, so are $f + g$ and $\max\{f, g\}$.
(iii) If f_n are lower semicontinuous, so its $f(x) = \sup_n f_n(x)$.
(iv) A function f is lower semicontinuos if and only if there is a monotone increasing sequence ϕ_n of continuous functions such that $f(x) = \lim \phi_n(x)$.

(i) ($\Rightarrow$) Let f be lsc, and fix $\lambda \in \mathbb{R}$ and $x_0 \in A_{f,\lambda} = \{f(x) > \lambda\}$. Then $f(x_0) > \lambda$ so $\epsilon = (f(x_0) - \lambda)/2 > 0$. By the definition of lim inf, exists $\delta > 0$ s.t. $\inf_{B_\delta(x_0)\setminus x_0} f(x) \geq \liminf_{x\to x_0} f(x) - \epsilon/2 \geq f(x_0) - \epsilon/2 > \lambda$. Hence, $f(x) > \lambda$ for $x \in B_\delta(x_0)$, ie $B_\delta(x_0) \subset A_{f,\lambda}$. Since λ and x_0 were arbitrary, $A_{f,\lambda}$ is open, $\forall\lambda$.

($\Leftarrow$) Let $A_{f,\lambda}$ be open for all λ, and x_0. Let $\epsilon > 0$. Since $A_{f,f(x_0)-\epsilon}$ is open and contains x_0, $f(x) \geq f(x_0) - \epsilon$ in some neighborhood of x_0, therefore $\liminf_{x\to x_0} \geq f(x_0)$. This is true for all ϵ, so $\liminf_{x\to x_0} f(x) \geq f(x_0)$.

(ii) Given $\epsilon > 0$, clearly $\inf_{B_\delta(y)\setminus y} (f(x) + g(x)) \geq \inf_{B_\delta(y)\setminus y} f(x) + \inf_{B_\delta(y)\setminus y} g(x)$, so taking limit as $\epsilon \to 0$ gives $\liminf_{x\to y}(f(x) + g(x)) \geq \liminf_{x\to y} f(x) + \liminf_{x\to y} g(x) \geq f(y) + g(y)$ since f and g are lsc at y.

Again, given $\epsilon > 0$, $\inf_{B_\delta(y)\setminus y} max\{f(x), g(x)\} \geq \inf_{B_\delta(y)\setminus y} f(x)$ since $max\{f, g\} \geq f$. Taking limit as $\epsilon \to 0$ gives $\liminf_{x\to y} \max\{f(x) + g(x)\} \geq \liminf_{x\to y} f(x) \geq f(y)$, since f is lsc at y. Analogously $\liminf_{x\to y} \max\{f(x) + g(x)\} \geq g(y)$, hence $\liminf_{x\to y} \max\{f(x) + g(x)\} \geq max\{f(y), g(y)\}$.

(iii) $A_{f,\lambda} = \cup_n A_{f_n,\lambda}$, since given x, if all $f_n(x) \leq \lambda$, $f(x) \leq \lambda$ too. $A_{f_n,\lambda}$'s are open, hence so is $A_{f,\lambda}$.

(iv) ($\Rightarrow$) Divide $\mathbb{R}$ in intervals $[\frac{k}{2^n}, \frac{k+1}{2^{n+1}})$ and let $f_n(x)$ be $infimum$ of f in $[\frac{k}{2^n} - \frac{1}{2^{n+2}}, \frac{k+1}{2^n} + \frac{1}{2^{n+2}})$ for all x in $[\frac{k}{2^n} + \frac{1}{2^{n+2}}, \frac{k+1}{2^n} - \frac{1}{2^{n+2}})$, with linear interpolation in between them. Using compactness of the closure of the interval and lower semicontinuity, one can easily show that the $infimum$ is not $-\infty$.

Now let $\phi_n = \max\{f_1, \ldots, f_n\}$. ϕ_n's are continuous, since so are f_n's. They are clearly increasing in n. Finally, $f_n(y) \leq f(y)$, so $\sup_n \phi_n(y) \leq f(y)$. On the other hand, given ϵ, $\{x : f(x) > f(y) - \epsilon\}$ is open, so $f(x) > f(y) - \epsilon$ in a neighborhood of y. Then, for a large enough n, $f_n(y) \geq \{\inf f(x)$ over a small interval of $y\} \geq f(y) - \epsilon$, giving $\sup \phi_n(y) \geq f(y) - \epsilon$. This is true for all ϵ, therefore $\sup \phi_n(y) = f(y)$.

($\Leftarrow$) $f(x) = \sup \phi_n(x)$ since the sequence is increasing, and $\phi(n)$'s are continuous, hence lsc, thus by (iii), f is lsc too. □

Problem 4:
(i) The set of points of continuity of f is a $\mathcal{G}_\delta$.
(ii) $\mathbb{Q}$ is not a $\mathcal{G}_\delta$.

(i) Let C be the set of continuity of f. f is continuous at y iff $\liminf_{x\to y} f(x) = \limsup_{x\to y} f(x)$, so

$$C^c = \{y : \liminf_{x\to y} f(x) < \limsup_{x\to y} f(x)\} = \cup_{p<q\in\mathbb{Q}}\{\{\liminf_{x\to y} f(x) \leq p\} \cap \{\limsup_{x\to y} f(x) \geq q\}\}.$$

Hence, it's enough to show that $\{\liminf_{x\to y} f(x) \leq p\}$ and $\{\limsup_{x\to y} f(x) \geq q\}$ are closed (this implies C^c is F_σ). But one can easily show that $\{\liminf_{x\to y} f(x) > p\}$ and $\{\limsup_{x\to y} f(x) < q\}$ are open, proving therefore that C is G_δ.

(ii) Suppose $\mathbb{Q} = \cap O_n$, O_n's open. Let $q_1, q_2, \ldots$ be an enumeration of the rationals, and consider $O'_n = O_n\setminus\{q_n\}$. Now $\bar{O'n} = \bar{O}_n = \mathbb{R}$, so Baire's category theorem implies $\cap O'_n$ is dense too, contradicting $\cap O'_n = \emptyset$. □

Problem 5: Prove that $\mathcal{M}$ is a σ-algebra and that μ is a measure on $\mathcal{M}$.

$\emptyset$ is countable, hence in $\mathcal{M}$. If $E \in \mathcal{M}$, it's either countable or co-countable, hence so is E^c. Finally, if $\{E_n\} \subseteq \mathcal{M}$, then either all of them are countable, hence so is their union; or at least one is co-countable, say E_m, so $(\cup E_n)^c \subseteq E_m^c$ is countable too. Therefore $\mathcal{M}$ is a σ-algebra.

Clearly μ is non-negative and $\mu(\emptyset) = 0$. Now if $E, E_n \in \mathcal{M}$ with $E \subset \cup E_n$, then either all of E_n are countable, in which case so is their union and therefore E, giving $\mu(E) \leq \sum \mu(E_n)$ (all of them0); or at least one E_n is

co-countable. In this case $\sum \mu(E_n) \geq 1$, whereas $\mu(E)$ is either 0 or 1, so again $\mu(E) \leq \sum \mu(E_n)$. Hence, μ is a measure on $\mathcal{M}$. □

Problem 6: Construct a Borel set E such that for every interval I, $0 < m(E \cap I) < m(I)$.

First note that neither E nor E^c can contain an open set, hence E needs to be nowhere dense. The strategy is to find inside each bounded rational interval $I_1, I_2, \ldots$ two disjoint nowhere dense Borel sets of positive measure. We start by claiming that any interval I has a nowhere dense subset A of positive measure. In deed, let $0 < \epsilon < m(I)/2$, and $q_1, q_2, \ldots$ the rationals in I. Let $A = I \backslash (\cap B_{\epsilon 2^{-n}}(q_n))$. Then $op\,(cl\,(A)) = \emptyset$ iff $(op\,(cl\,(A)))^c = \mathbb{R}$. But

$$(op\,(cl\,(A)))^c = cl\,((cl\,(A))^c) = cl\,(op\,(A^c)) = cl\,(op\,(I^c \cap (\cap B_{\epsilon 2^{-n}}(q_n)))) = cl\,(op\,(I^c) \cap (\cap B_{\epsilon 2^{-n}}(q_n))) \supseteq cl\,(\mathbb{Q}) = \mathbb{R},$$

so A is nowhere dense, Borel, and $m(A) \geq m(I) - \sum m(B_{\epsilon 2^{-n}}(q_n)) \geq m(I)/2$.

Now define inductively S_n, E_n, F_n as follows:

$E_1, F_1 \subset I_1$ disjoint, Borel, nowhere dense, of positive measure (pick E_1, then $I_1 \backslash E_1$ will have some open interval, so pick F_1 there).

$S_n = (E_1 \cup \ldots E_{n-1}) \cup (F_1 \cup \ldots F_{n-1})$ for $n \geq 2$.

Given S_n, define $E_n, F_n \subset I_n \backslash S_n$ disjoint, Borel, nowhere dense, of positive measure, and this can be done because S_n is nowhere dense.

Let $E = \cup_n E_n$, $F = \cup_n F_n$, both disjoint Borel sets. If I is any non-empty interval, it contains some I_n, so

$$m(I \cap E) \geq m(I_n \cap E_n) = m(E_n) > 0,$$

$$m(I) \geq m(I \cap E) + m(I \cap F) \geq m(I \cap E) + m(I_n \cap F_n) > m(I \cap E).$$

□

Problem 7: Show that there is a support of every Borel measure. Show that every compact set is the support of some Borel measure.

Let $\mathcal{F} = \{(p,q) : p, q \in \mathbb{Q}, \mu((p,q)) = 0\}$. If $\mathcal{F} = \emptyset$, then $K = [0,1]$ because each open interval has positive measure. Otherwise, we claim that $K = O^c$, where $O = (\cup_{\mathcal{F}}(p,q))^c$. Clearly K is compact and $m(K) = 1 - m(O) = 0$. If the compact set $H \subset K$ is a proper subset, $K \backslash H$ contains a rational interval (p,q), and $(p,q) \subset K = O^c$, so (p,q) is not in $\mathcal{F}$. This implies that $m(p,q) > 0$, therefore $m(H) \leq 1 - m(p,q) < 1$.

If $K \subset [0,1]$ is compact & non-empty, it has a dense sequence $x_1, x_2, \ldots$, possibly finite. If it is a finite sequence $x_1, \ldots, x_N$, just call $x_n = x_N$ for $n > N$. Define μ by on the Borel σ-algebra via $\mu(S) := \sum 2^{-n} \delta_{x_n}(S)$. Then $\mu([0,1]) = 1$, and its support is K. □

Problem 8: Let V be an open subset in $\mathbb{R}$ and μ a finite Radon measure. Define $f(x) = \mu(V + x)$, does f have to be continuous, lower semicontinuous or upper semicontinuous?

f is lower semicontinuous: Let $x_0 \in A_{f,\lambda}$ and $\epsilon = f(x_0) - \lambda$. μ is a Radon measure, so there exists a compact $K \subset (V + x_0)$ with $\mu((V + x_0) \backslash K) < \epsilon/2$. Since $V + x_0$ is open, $\delta = dist\,(K, (V + x_0)^c) > 0$.
We claim that $B_{\delta/2}(x_0) \subset A_{f,\lambda}$. In deed, $||x - x_0|| < \delta/2$ implies $K \subset (V + x) \cap (V + x_0)$, since if $k \in K$,

$$d(k, (V+x)^c) = d(k - (x - x_0), (V + x_0)^c) \geq d(k, (V + x_0)^c) - d(k, k - (x - x_0)) \geq \delta - \delta/2 > 0 \Longrightarrow k \in (V + x).$$

This means that $f(x) = \mu(V + x) \geq \mu(K) = \mu(V + x) - \mu((V + x_0) \backslash K) \geq (\lambda + \epsilon) - \epsilon/2 > \lambda$, hence $x \in A_{f,\lambda}$.
This proves that $A_{f,\lambda}$ is open for all λ.

f is not necessarily upper semicontinuous: For $\mu = \delta_0$, $V = (0,1)$: f is 0 on $(-\infty, 0] \cap [1, \infty)$ and 1 on $(0,1)$. □

Problem 9: Construct a function such that each set $\{f(x) = \alpha\}$ is measurable for any $\alpha \in \mathbb{R}$, but the set $\{f(x) > 0\}$ is not measurable.

Let $P \in [0, 1)$ be the non-measurable set constructed by picking exactly one element of each of the equivalence classes of the relation $x \sim y \iff x - y \in \mathbb{Q}$. Define f by

$$f(x) = \begin{cases} x & x \in P \\ -|x| & x \notin P. \end{cases}$$

$\{f(x) = \alpha\} \subseteq \{-\alpha, +\alpha\}$ is discrete, hence measurable, whereas $\{f(x) > 0\} = P$, which is not measurable. $\square$